

SIH – Smart Scan Strategy Crash Course

RF Detection • Adaptive Scanning • Bandit Scheduling • Detection Theory

1. Simulated Environment & Detection Model

The simulated environment contains hidden ground truth.

$Z_{it} = 1$ if an emitter is active on frequency i at time t ; otherwise $Z_{it} = 0$.

$Y_{it} \in \{0, 1\}$, where $Y_{it} = 1$ means detection is declared.

Quantity	Meaning	Who can see it?
Z_{it}	True emitter activity in the simulation	Evaluator / Simulator
Y_{it}	Receiver's hit or alarm	Scheduler
A_t	Receiver's decision / next band choice	Scheduler
Reward r_t	Numerical score per decision	Learning algorithm

2. RF & Detection Essentials

Frequency

- Rate of electromagnetic oscillation.
- Determines what the receiver can monitor at a given moment.

Bandwidth

- Instantaneous bandwidth: frequency range the receiver can monitor at one time.
- Source zone / total search range: total frequency range that needs to be searched.

If N frequency bins exist and the receiver can handle K bins at a time: Time to cover $\approx N/K$.

3. Signal-to-Noise Ratio & Sensitivity

$$\text{SNR} = P_{\text{signal}} / P_{\text{noise}}$$

$$\text{SNR}_{\text{dB}} = P_{\text{signal,dB}} - P_{\text{noise,dB}}$$

$$P_N \approx kTBF$$

where k = Boltzmann constant, T = receiver temperature, B = bandwidth in Hz, and F = receiver noise factor.

$$P_N(\text{dBm}) \approx -174 + 10 \log_{10}(B) + NF$$

Example: $B = 1 \text{ MHz}$ and $NF = 5 \text{ dB}$:

$$P_N = -174 + 60 + 5 = -109 \text{ dBm}$$

This represents the approximate receiver noise floor. Sensitivity should be specified together with the required probability of detection, false-alarm probability, bandwidth, and dwell time.

- Example requirement: $P_D \geq 90\%$ and $P_{FA} \leq 0.01$.

4. Four Possible Detection Outcomes

Ground Truth	Receiver Decision	Result
Signal Active	Detection	True Positive
Signal Active	No Detection	Miss
No Signal	Detection	False Alarm
No Signal	No Detection	True Negative

$$P_D = P(Y = 1 | Z = 1)$$

$$P_{FA} = P(Y = 1 | Z = 0)$$

$$P_M = P(Y = 0 | Z = 1) = 1 - P_D$$

5. Threshold & ROC Trade-off

Let γ be the detection threshold.

- Lower $\gamma \rightarrow$ higher detection probability but more false alarms.
- Higher $\gamma \rightarrow$ fewer false alarms but more misses.

The ROC curve shows the trade-off between P_D (probability of detection) and P_{FA} (probability of false alarm). An ideal detector achieves high P_D while maintaining low P_{FA} .

6. From Sweep to Smart Scan

Sequential Scan

$$f_1 \rightarrow f_2 \rightarrow \dots \rightarrow f_N$$

$$T_{\text{sweep}} \approx N(t_d + t_s)$$

where t_d = dwell time and t_s = switching time.

7. Scan Metrics

Metric	Definition
Revisit Time	Time between two visits to the same band
Dwell Time	Time spent observing a selected band
Switching Time	Time lost while changing frequency
Intercept Time	Time from activity beginning to first detection
Intercept Rate	Detected activity / total activity
P_D	Detection probability when observing an active band
P_{FA}	Alarm probability when observing an empty band
Time Prediction Error	Difference between predicted and actual activity time

$$t_I = t_{\text{first valid detection}} - t_{\text{emission start}}$$

$$\text{Average Intercept Time} = (1/N_D) \sum \text{Intercept Time}_i$$

8. Adaptive Scanning as a Bandit Problem

- Each frequency band → Arm
- Choosing a band → Action
- Feedback / reward → Learning signal

The scheduler must balance:

$$\text{Exploration} \leftrightarrow \text{Exploitation}$$

Exploitation

- Focus on known / high-value bands.
- Risk: locking onto a band and missing new activity.

Exploration

- Search less-known or uncertain bands.
- Helps discover new activity when time is limited.

9. RL, Periodic Emitters & System Architecture

- Synthetic RF Environment
- Truth Generator: $Z(t)$
- Receiver Model: limited instantaneous bandwidth; dwell and return time; noise/SNR model; P_D/P_{FA} detection model
- Scheduler: Round Robin, UCB, Thompson Sampling, Contextual Bandit / policy
- Metric Engine: P_D , P_{FA} , Intercept Rate, Average Intercept Time, Prediction Time Error, Reward / Cost
- Plot & Comparison Report

10. Cost-Aware Reward Design

$$R_t = w_H \cdot \text{Hit} - w_{FA} \cdot \text{FalseAlarm} - w_T \cdot \text{TimeCost} - w_S \cdot \text{SwitchCost} + w_N \cdot \text{NovelIntercept}$$

A novel intercept is the first successful detection of a new activity episode.

11. UCB – Upper Confidence Bound

- Uses estimated reward plus an uncertainty bonus.
- Provides explainable and deterministic exploration.

$$\text{Score}_i = \hat{Q}_i + c \sqrt{[\ln(t)] / (N_i + 1)}$$

$\hat{Q}_i$ = estimated utility of band i ; t = total decisions/time; N_i = visits to band i ; c = exploration-strength parameter.

12. Thompson Sampling

Thompson Sampling samples plausible reward values and selects the band with the best sampled value.

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$$

- Valid hit: $\alpha_i \leftarrow \alpha_i + 1$
- Miss: $\beta_i \leftarrow \beta_i + 1$

13. Contextual Bandit

A contextual bandit selects a band using current parameters/features such as age, recent hits, and predicted phase.

- Useful for time-varying / non-stationary activity.
- Decisions can depend on the current context.

14. RL / MDP

Reinforcement Learning / Markov Decision Process learns actions based on current state, available actions, resulting rewards, and future states.

The objective is to learn a policy where the current choice can affect future states and opportunities.

15. Spectrogram vs Truth Map

- Spectrogram = Frequency × Time picture.
- Horizontal axis → Time
- Vertical axis → Frequency
- Colour intensity → Measured power

Bright regions may represent active signals, noise, or interference.

A truth map represents actual simulated emitter activity, while a spectrogram represents what the receiver observes.

16. Detection Theory

$$H_0: x(t) = n(t) \rightarrow \text{No signal, noise only}$$

$$H_1: x(t) = s(t) + n(t) \rightarrow \text{Signal + noise}$$

The receiver obtains a measurement X and calculates a test statistic $T(X)$, which is compared against threshold γ .

$$T(X) \geq \gamma \rightarrow \text{Declare Detection}$$

$$T(X) < \gamma \rightarrow \text{Declare No Detection}$$

17. Scheduling Algorithms

Round Robin: Each band receives an equal opportunity to be scanned.

UCB: Selects bands using estimated utility plus an uncertainty bonus.

Thompson Sampling: Samples plausible band utilities and selects the best sampled value.

Contextual Bandit: Uses additional context and current conditions to select the next band.

RL / MDP: Learns a policy considering how current actions affect future states and rewards.

Core Objective

Maximize useful detections while minimizing false alarms, scan time and switching cost.

Maintain the balance: Exploration $\leftrightarrow$ Exploitation

Evaluate using P_D, P_FA, Intercept Rate, Average Intercept Time, Prediction Error, and Reward/Cost.